



## S1.3 Electron configurations

Structure 1 · Models of the particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

### Understand the concept

Electrons occupy quantised energy levels (shells) which are divided into sub-shells (s, p, d, f), each containing orbitals that hold a maximum of two electrons with opposite spins. Sub-shells fill in order of increasing energy (1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p ...), following the Aufbau principle, and within a sub-shell electrons occupy separate orbitals singly before pairing (Hund's rule). Configurations can be written in full (e.g.  $1s^2 2s^2 2p^6$ ) or using noble-gas core notation; transition metals show characteristic exceptions (e.g. 4s fills before 3d, but electrons are removed from 4s first when forming ions, and Cr and Cu have anomalous configurations).

### Worked example: follow the reasoning

Deduce the full electron configuration of an iron(III) ion,  $Fe^{3+}$  ( $Z = 26$ ).

Step 1: Neutral Fe (26 electrons):  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$ .

Step 2: To form  $Fe^{3+}$ , remove 3 electrons. Electrons are always removed from the highest energy/outmost sub-shell first, which for transition metal ions is 4s before 3d.

Step 3: Remove both 4s electrons first, then one 3d electron:  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$ .

Answer:  $Fe^{3+}$ :  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$  (23 electrons).

### Quick check · multiple choice

What is the electron configuration of a chloride ion,  $Cl^-$ ?

A  $1s^2 2s^2 2p^6 3s^2 3p^5$

B  $1s^2 2s^2 2p^6 3s^2 3p^6$

C  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

D  $1s^2 2s^2 2p^6 3s^2 3p^4$

Write your choice: \_\_\_\_\_

### Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 (a) Write the full electron configuration of a ground-state vanadium atom, V ( $Z = 23$ ). [1] [5]  
(b) Deduce the electron configuration of the  $V^{3+}$  ion. [1]  
(c) Explain, in terms of electron configuration, why transition metals commonly form coloured ions/compounds while ions of s-block metals such as  $Na^+$  do not. [3]

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Final answer: \_\_\_\_\_



## Answers and explanation

### Question 1 [5]

(a)  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$ . [1]

(b) Remove the two 4s electrons and one 3d electron:  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2$ . [1]

(c) Transition metal ions have partially filled d sub-shells; the d orbitals split into two energy sub-levels in the presence of ligands, and electrons can be promoted between them by absorbing a specific wavelength of visible light, so the complementary colour is transmitted/reflected. s-block ions such as  $Na^+$  have no d electrons (empty or full sub-shells only), so no such d-d electron transition is possible and no colour arises. [3]

### Quick check answer

B.  $Cl^-$  has 18 electrons ( $17 + 1$ ), giving the stable noble-gas configuration of argon,  $1s^2 2s^2 2p^6 3s^2 3p^6$ .

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.